AAI3008
Large Language Models
Introduction

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Module Roadmap

Data & Representation

- Tokenize
- Embedding

Model Architecture

- Transformer
- Recurrent Neural Network
- LLM Adaptation

Tasks & Evaluation

- Question & Answering
- Machine Translation
- Chatbot

System & Applications

- Prompt Engineering
- Retrieval Augmented Generation (RAG)
- Spoken Language Processing
- Multimodal LLM

Outline

- Natural Language Processing Tasks
  - **Word Level, Sentence Level, Article Level …**
  - **NLTK**

- Recurrent Neural Network (RNN)
  - For sequence processing
  - Application: POS Tagging

NLTK (Natural Language Toolkit)

- A Python library for processing human language data.
  - Developed by Steven Bird and Edward Loper in 2001.
  - Provides tools for tasks like tokenization, tagging, parsing, and semantic reasoning.
  - Includes a wide range of corpora and lexical resources.
- Install NLTK library
  - pip install nltk

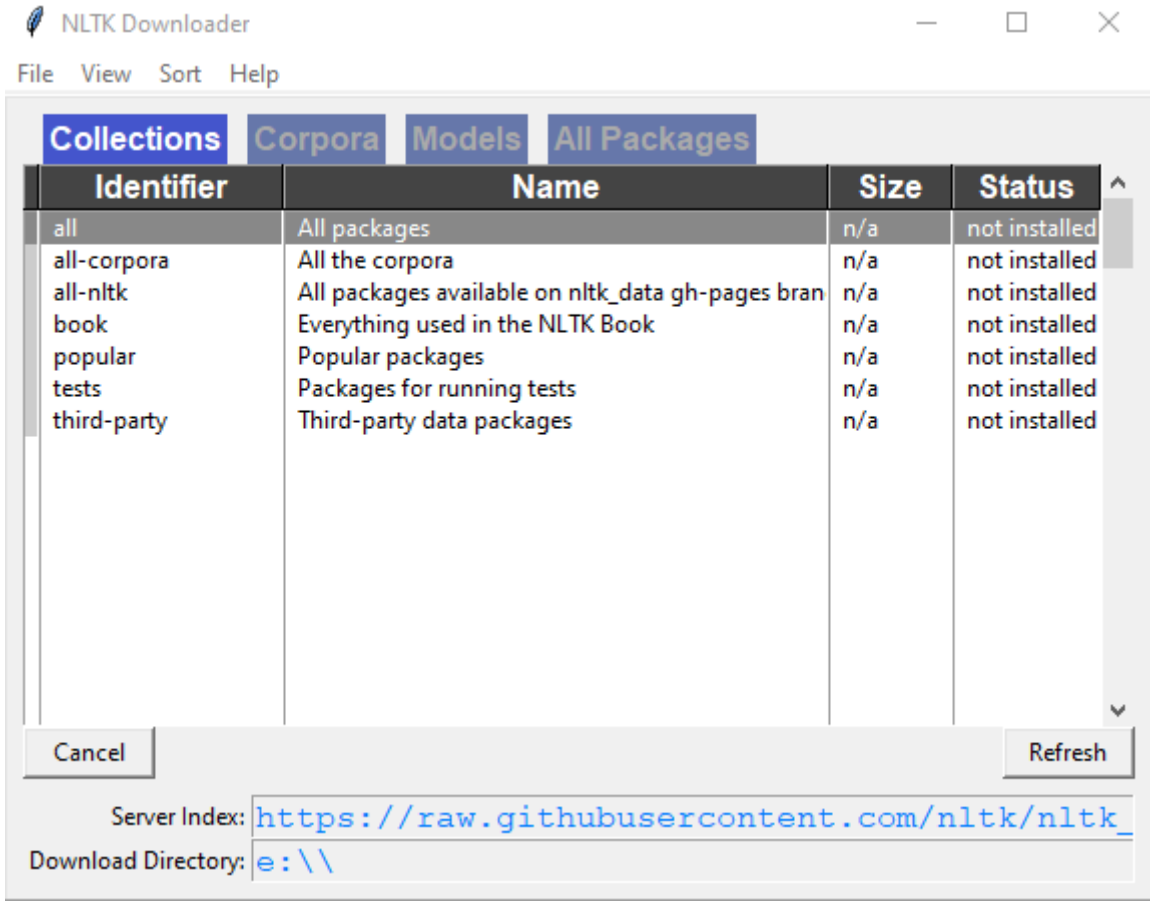
NLTK

- **Core Functionalities[1]**
  - Lexical Analysis
    - Part-of-Speech (POS) tagging.
    - Stemming and Lemmatization.
  - Syntactic Parsing
    - Context-Free Grammar (CFG).
    - Dependency Parsing.
  - Text Preprocessing
    - Tokenization (splitting text into words/sentences).
    - Stopword removal.
  - Semantic Analysis
    - WordNet integration for word relationships.

1. Sumamrized by ChatGPT
2. https://www.cybrosys.com/blog/what-are-the-key-features-of-the-natural-language-toolkit-nltk-in-python

Steps to Install NLTK

- Download NLTK datasets:
  - import nltk
  - nltk.download()



Spacy

- pip install spacy
- python -m spacy download en_core_web_sm

- An open-source NLP library designed for efficient, production-ready natural language processing.
- It provides fast and accurate implementations of core NLP tasks such as tokenization, part-of-speech tagging, named entity recognition, and syntactic parsing (both dependency and constituency-related representations).

Review and Discussion

- What does it mean to understand a piece of text?
  - I can identify typos.
  - I can answer questions about the text:
    - Who, when, what, why, and how.
    - Phrase Chunking: Identifying noun phrases.
    - Coreference Resolution: Determining who "she" refers to.
    - Discourse Relationship Recognition: Understanding why "she" stays at home.
  - Summarizing the text.

Review and Discussion

- How to understand it?
  - Try to find the answers in this module
  - Words → Phrases → Sentences
    - Divide and conquer
  - Machine Learning
  - Attention is all you need but data is king

Impact of Large Language Models

- Emergent Abilities
  - Large Language Models (LLMs) such as GPT-4 exhibit emergent capabilities
  - Perform tasks like POS tagging **without explicit task-specific training**
  - This reduces reliance on traditional supervised learning pipelines and feature engineering

Perform part of speech tagging for the following sentence: "John bought 3 apples for $5.99"

Here is the **Part-of-Speech (POS) tagging** for the sentence **"John bought 3 apples for $5.99"**:

1. **John**: Proper Noun (NNP)
   - Represents a specific person's name.
2. **bought**: Verb, Past Tense (VBD)
   - Describes an action (past tense of "buy").
3. **3**: Cardinal Number (CD)
   - Indicates a specific quantity.
4. **apples**: Noun, Plural (NNS)
   - Refers to multiple objects (plural form of "apple").
5. **for**: Preposition (IN)
   - Indicates purpose or exchange.
6. **$**: Symbol ($)
   - Denotes currency.
7. **5.99**: Cardinal Number (CD)

Large Language Models

- EMNLP2023

Keynote Talk: Academic NLP research in the Age of LLMs: Nothing but blue skies!

Christopher D. Manning
Stanford University

Sunday, December 10 – Time: **14:00 - 15:00** – Room: **East & Center**

Abstract: There has been a certain amount of handwringing by students – and faculty –about the prospects for academic research in the Age of Large Language Models (LLMs). Now, there is a need for universities and governments to do more to support computational research. And it is indeed the case that any major inflection point in a research field does change the best questions to concentrate your time on. But a reorientation of research is healthy, coming off a decade when students in academia had often drifted into playing the Kaggle game rather than thinking hard about fundamental ideas. A time of transformation is principally a time of opportunity: All sorts of new and exciting research questions, either not explored or only explored in very different contexts, come to the fore. These questions provide compelling opportunities for fresh and exciting work. I see many examples of this sort of work already happening and encourage many more. I will illustrate with two pieces of work from my own students: the Backpack, which explores an alternative, more interpretable architecture than the Transformer (by John Hewitt and colleagues) and Direct Preference Optimization, which explores an alternative to the Proximal Policy Optimization normally used for steering LLMs in the Reinforcement Learning from Human Feedback (RLHF) phase (by Eric Mitchell and colleagues).

Impact of Large Language Models

- New Research Directions
  - Multilingual and Low-Resource Scenarios
  - Cross-Modal Integration
    - Video understanding
  - Efficient Adaptation
    - Prompt engineering
    - Lightweight fine-tuning (e.g., LoRA, Adapters)
    - Retrieval-augmented generation

Outline

- Natural Language Processing Tasks
  - Applications
  - Word Level, Sentence Level, Article Level …
- **Recurrent Neural Network (RNN)**
  - For sequence processing
  - Application: POS Tagging

Sequential Data

- Task: Classify the sentiment category of a sentence
- Dataset
  - Today's weather is not bad. Positive
  - The weather is bad. Negative
- Model
  - Feedforward Neural Network
- Output
  - Sentiment category: -1, 1
- Input
  - Word? Word→Vector

	Word	One-hot Encoding
1	Today	[1,0,0,0,0,0,0]
2	's	[0,1,0,0,0,0,0]
3	Weather	[0,0,1,0,0,0,0]
4	Is	[0,0,0,1,0,0,0]
5	Not	[0,0,0,0,1,0,0]
6	Bad	[0,0,0,0,0,1,0]
7	the	[0,0,0,0,0,0,1]

https://text2data.com/Demo

Feedforward Neural Network

$$h = \mathrm{ReLU}(W_1 \mathrm{X} + b_1)$$

$$\mathrm{Y} = \mathrm{Softmax}(W_2 h + b_2)$$

$$\mathrm{Loss} = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

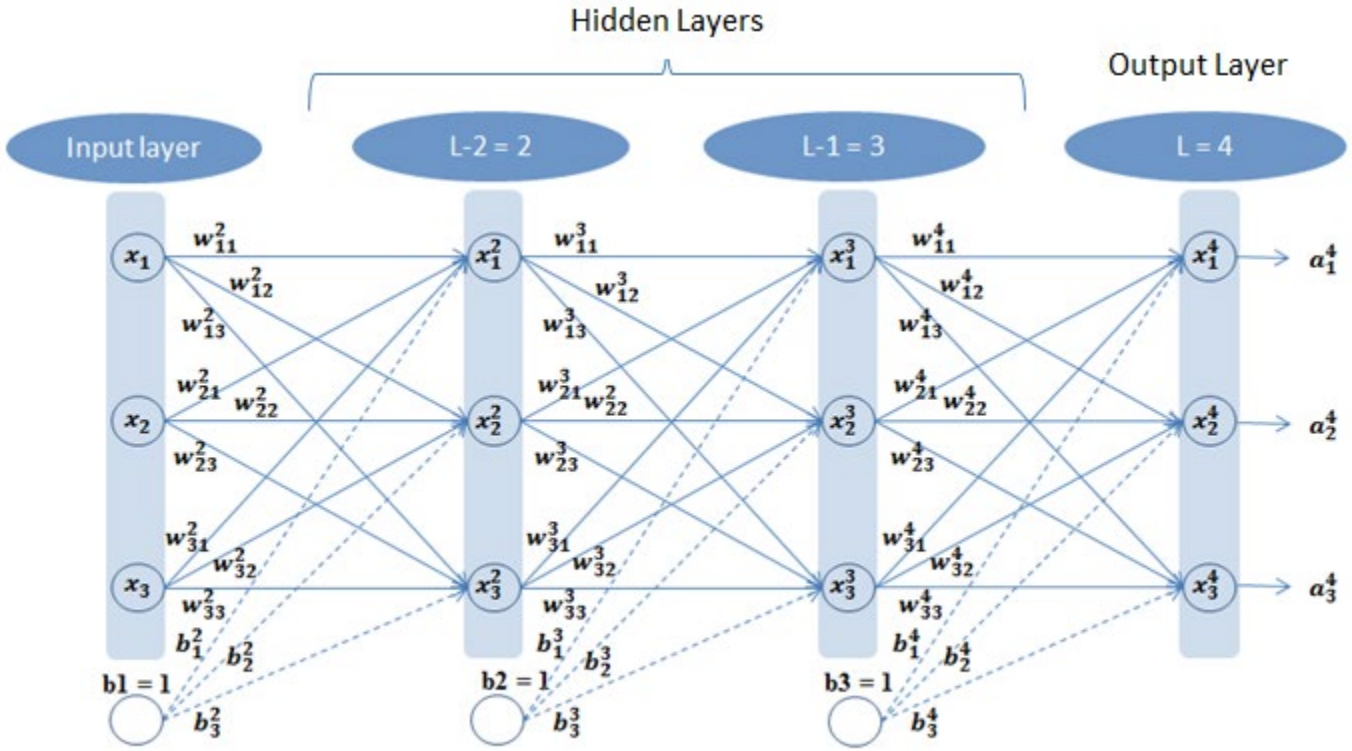


Figure 1

https://whyml.wordpress.com/2017/04/24/back-propagation-learning-for-multi-layer-feedforward-neural-network/

Sequential Data

- Variable-Length Text vs. Fixed-Length Input
  - Multiple words have to be mapped into a single fixed-dimensional vector
  - Summation and Average
    - Sum all word vectors and divided by word numbers
    - Problem?
      - *Good, not bad*
      - *Bad, not good*
      - No context information

	Word	One-hot Encoding
1	Today	[1,0,0,0,0,0,0]
2	's	[0,1,0,0,0,0,0]
3	Weather	[0,0,1,0,0,0,0]
4	Is	[0,0,0,1,0,0,0]
5	Not	[0,0,0,0,1,0,0]
6	Bad	[0,0,0,0,0,1,0]
7	the	[0,0,0,0,0,0,1]

Sequential Data

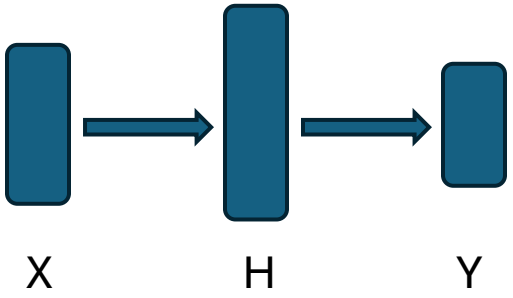
- Variable-Length Text vs. Fixed-Length Input
  - Concatenation and Padding
  - Concatenate all words to form a long vector.

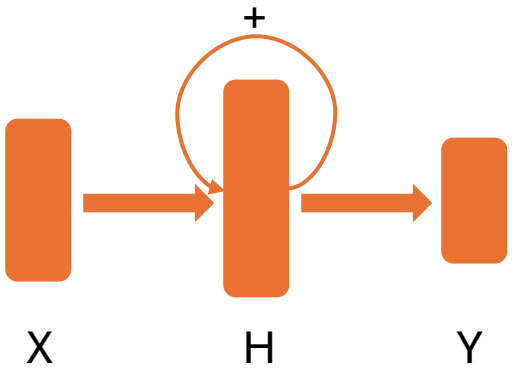
  - Problem?
    - High dimension
      - 7 times number of words
    - A lot of zeros
    - The meaning of each dimension is different

	Word	One-hot Encoding
1	Today	[1,0,0,0,0,0,0]
2	's	[0,1,0,0,0,0,0]
3	Weather	[0,0,1,0,0,0,0]
4	Is	[0,0,0,1,0,0,0]
5	Not	[0,0,0,0,1,0,0]
6	Bad	[0,0,0,0,0,1,0]
7	the	[0,0,0,0,0,0,1]

Recurrent Neural Networks (RNNs)

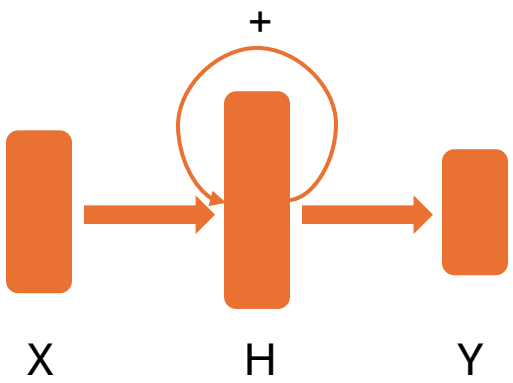
- Process the inputs one by one
- Remember the history

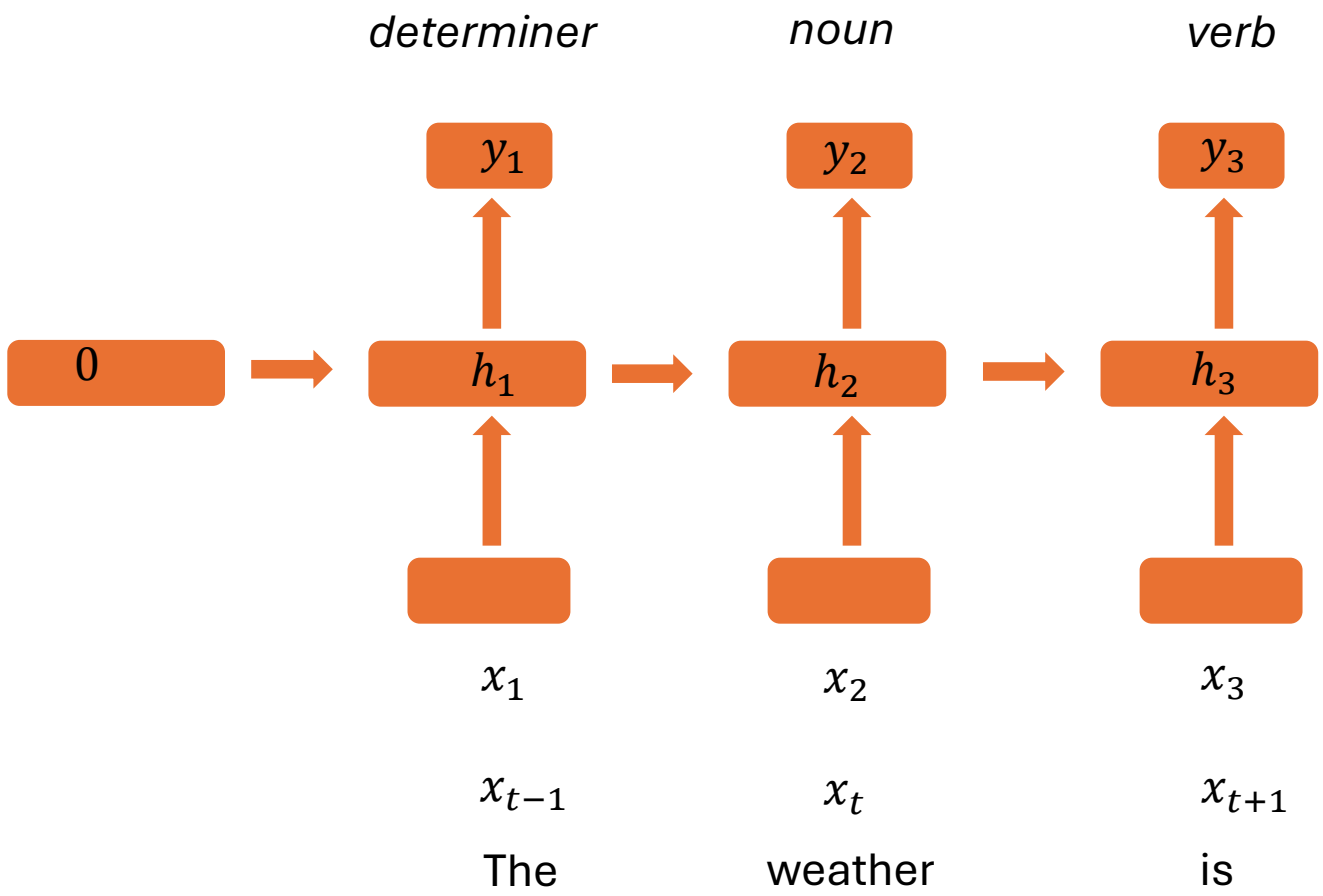




https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/

RNN Unfold



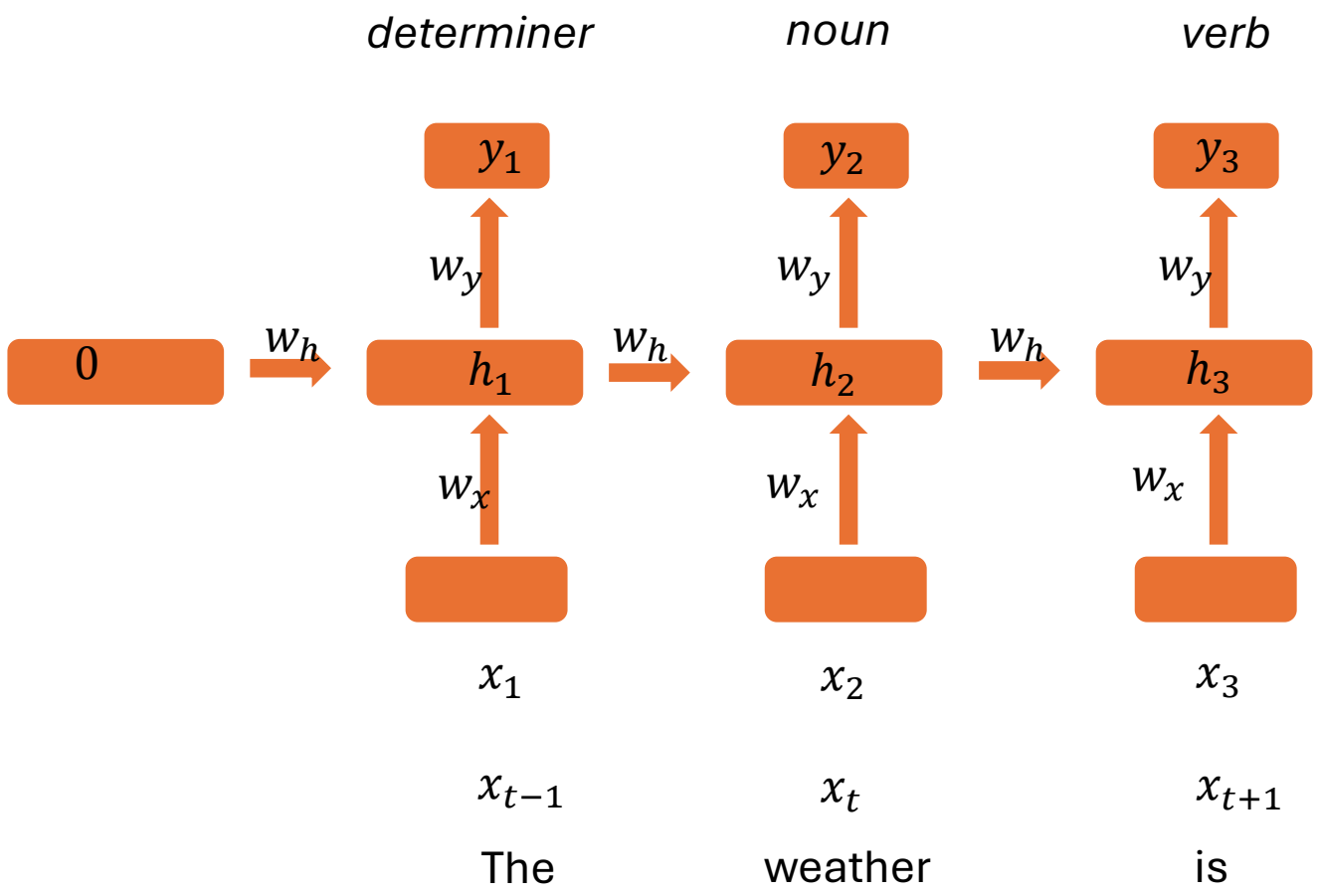


- Time Step t
  - A word
  - A segment of audio
  - Stock price of a day

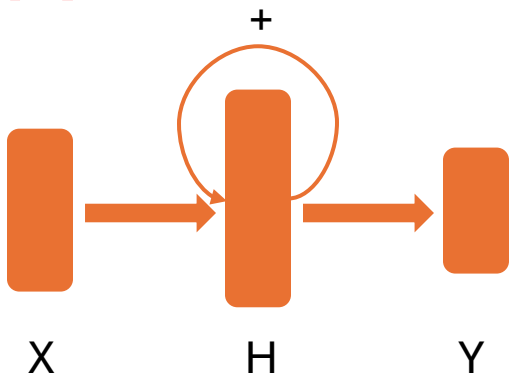
https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/

Forward Computation Process

- $h_t = f(W_h h_{t-1} + W_x x_t + b)$
- $y_t = g(W_y h_t + b_y)$
  - $\mathrm{Y} = \mathrm{Softmax}(W_2 h + b_2)$
- Time Step t
  - A word
  - A segment of audio
  - Stock price of a day



RNN



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- Time Step t
  - a word
  - A segment of audio
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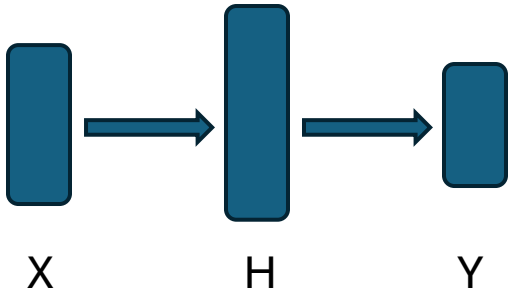

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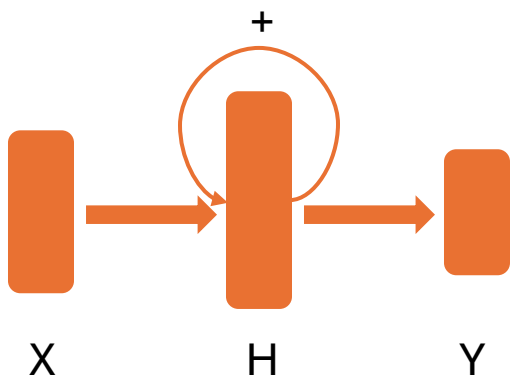
RNN

- Recurrence
  - Hidden state h_t depends on the current input (x_t) and **the previous state (h_{t-1})**
  - h_t stores information from current and past time steps
- Context information
- Variant length input

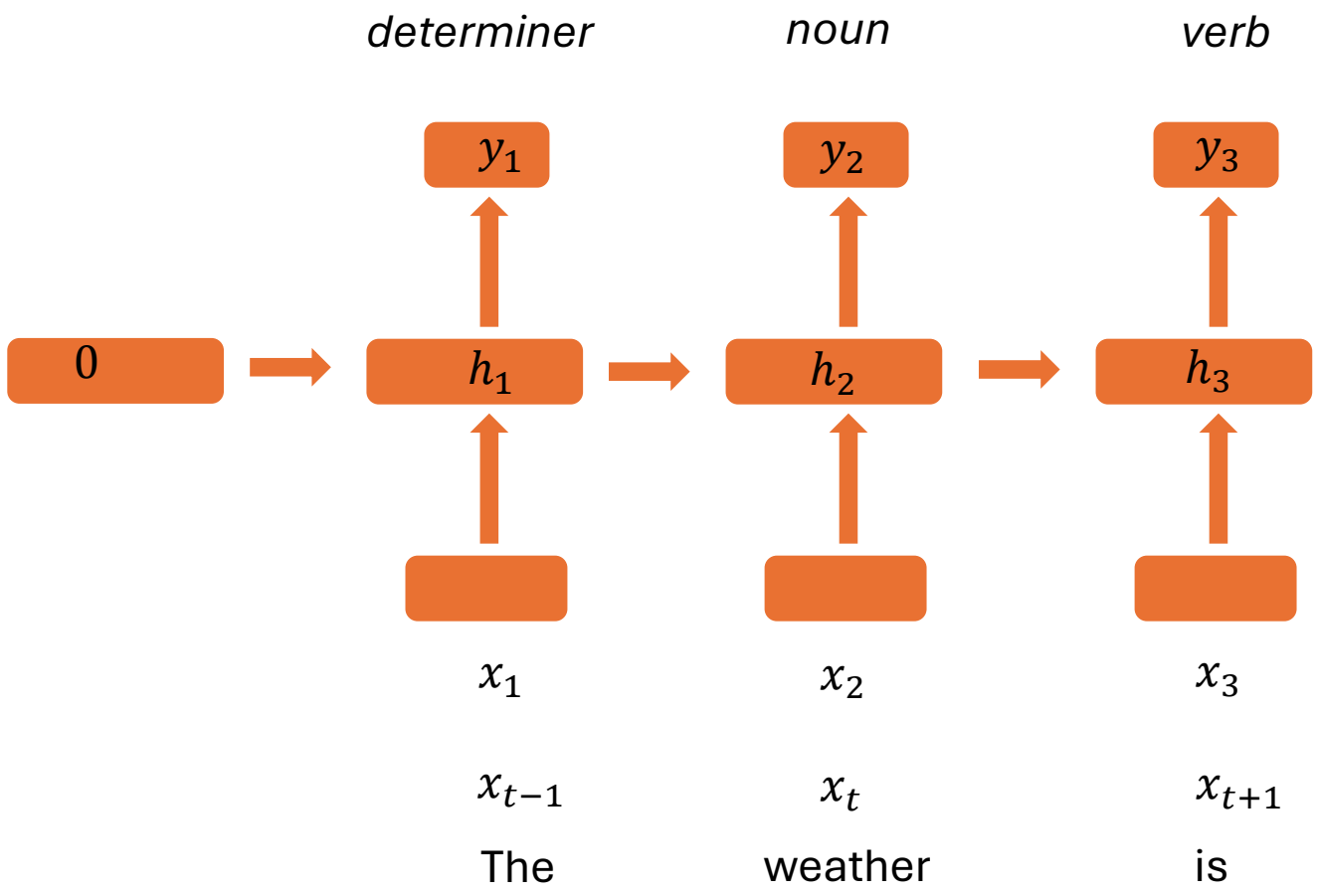


- $h_t = f(W_h h_{t-1} + W_x x_t + b)$
- $y_t = g(W_y h_t + b_y)$
  - $\mathrm{Y} = \mathrm{Softmax}(W_2 h + b_2)$
- Time Step t
  - t is a word
    - A segment of audio
    - Stock price of a day

https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/

Types Of Recurrent Neural Networks

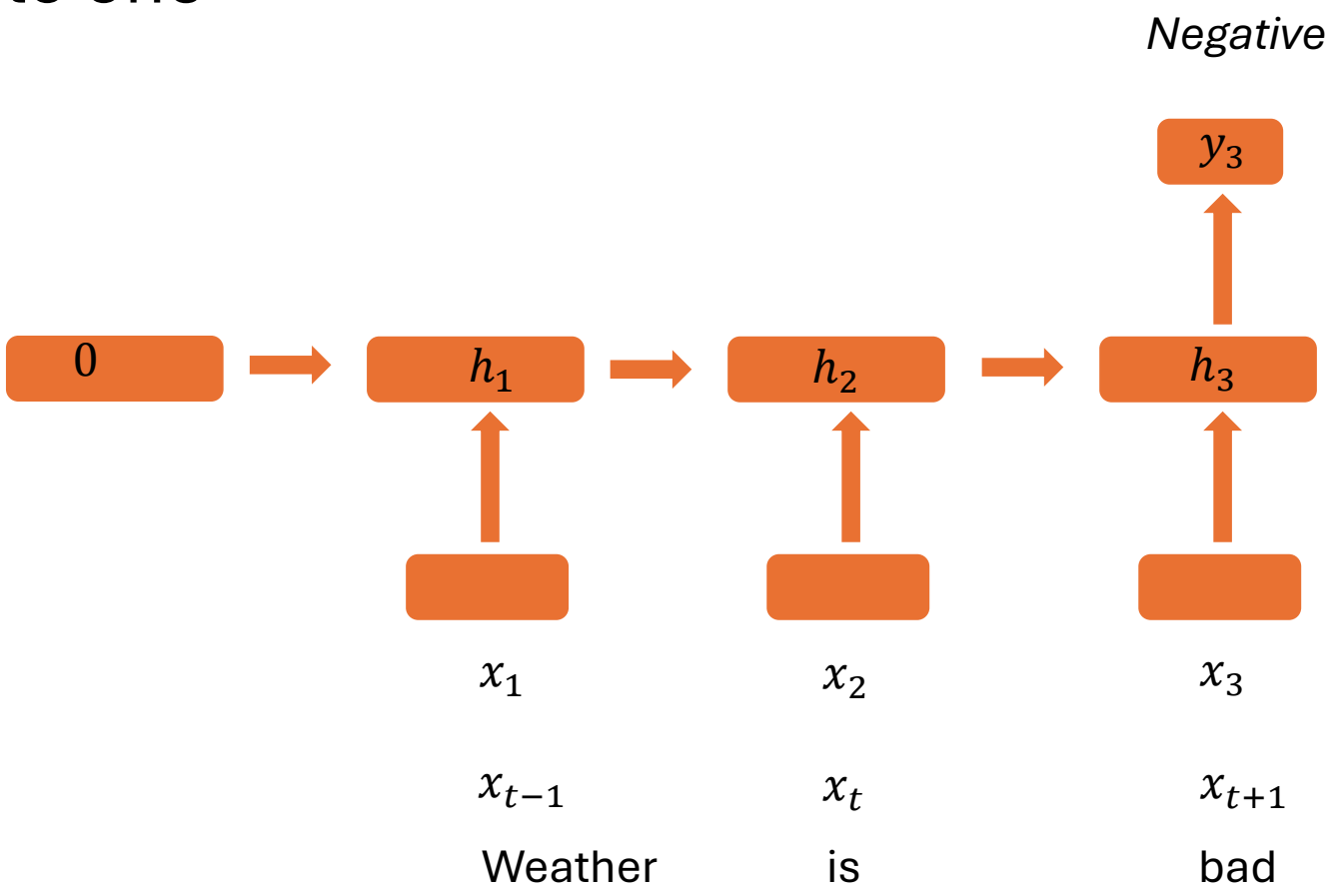
- Many to many



https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/

Types Of Recurrent Neural Networks

- Many to one



Types Of Recurrent Neural Networks

- One to one
- One to many
- https://www.geeksforgeeks.org/introduction-to-recurrent-neural-network/

Backpropagation Through Time (BPTT)

- **Vanishing Gradient**
  - Gradients diminish as they pass through each time step, leading to minimal weight updates.
  - This limits the RNN's ability to learn long-term dependencies, which is crucial for tasks like language translation.

  - Basic gradient descent update rule

$$w = w - \lambda \nabla_w \mathcal{L}$$

https://www.geeksforgeeks.org/vanishing-and-exploding-gradients-problems-in-deep-learning/

Backpropagation Through Time (BPTT)

- **Vanishing Gradient**
  - Gradients diminish as they pass through each time step, leading to minimal weight updates.
  - This limits the RNN's ability to learn long-term dependencies, which is crucial for tasks like language translation.
- **Exploding Gradient**
  - Sometimes, gradients grow uncontrollably, causing excessively large weight updates that destabilize training.
  - Gradient clipping is a common technique to manage this issue.

https://www.geeksforgeeks.org/vanishing-and-exploding-gradients-problems-in-deep-learning/

RNN Variants

- **Long Short-Term Memory (LSTM)**
  - Designed to overcome vanishing gradients.
  - Includes gates (input, forget, output) to control information flow.
- **Gated Recurrent Unit (GRU)**
  - Simplified version of LSTM with fewer gates.

Applications of RNNs

- Natural Language Processing:
  - Language modeling
  - Machine translation
- Speech Recognition:
  - Audio-to-text systems
- Time Series Analysis:
  - Stock price prediction

POS tagging

U.K. JJ B-NP
base NN I-NP
rates NNS I-NP
are VBP B-VP
at IN B-PP
their PRP$ B-NP
highest JJS I-NP
level NN I-NP
in IN B-PP
eight CD B-NP
years NNS I-NP
. . O

https://www.cnts.ua.ac.be/conll2000/chunking/
https://www.ling.upenn.edu/courses/Fall_2003/ling001/penn_treebank_pos.html

Tag	Description	Examples
CC	Coordinating conjunction	and, but, or, so
CD	Cardinal number	one, two, 100, 2023
DT	Determiner	the, a, an, this
EX	Existential there	there (in "there is")
FW	Foreign word	en route, doppelgänger
IN	Preposition or subordinating conjunction	in, on, by, because
JJ	**Adjective**	big, green, amazing
JJR	Adjective, comparative	bigger, better, faster
JJS	Adjective, superlative	biggest, best, fastest
LS	List item marker	1., 2., A., B.
MD	Modal	can, should, will
NN	**Noun**, singular or mass	dog, car, freedom
NNS	Noun, plural	dogs, cars, freedoms
NNP	Proper noun, singular	John, London, Apple
NNPS	Proper noun, plural	Americans, Rockies
PDT	Predeterminer	all, both, half
POS	Possessive ending	's (as in John's)
PRP	Personal pronoun	I, you, he, she, it, we, they
PRP$	Possessive pronoun	my, your, his, her, its, our, their
RB	**Adverb**	quickly, often, very
RBR	Adverb, comparative	faster, better
RBS	Adverb, superlative	fastest, best
RP	Particle	up, off, out (as in "pick up")
SYM	Symbol	$, %, &, +
TO	to (infinitive marker)	to (as in "to go")
UH	Interjection	oh, wow, hey
VB	**Verb**, base form	run, eat, go
VBD	Verb, past tense	ran, ate, went
VBG	Verb, gerund/present participle	running, eating, going
VBN	Verb, past participle	run, eaten, gone
VBP	Verb, non-3rd person singular present	run, eat (I run, you eat)
VBZ	Verb, 3rd person singular present	runs, eats (He runs, she eats)
WDT	Wh-determiner	which, that
WP	Wh-pronoun	who, what, whom
WP$	Possessive wh-pronoun	whose
WRB	Wh-adverb	where, when, why, how

POS tagging

- Lab 1: Given a sentence, predict the POS class of each word
- What problem is this? Classification or Regression?
- Which type of RNN should be used?
    1. Many to one
    2. Many to many
- What are the input and output?
- How to get the data?
- How to evaluate the model?

Summary

- Feedforward Neural Network
  - Cannot handle variant length inputs
  - No context information in sequences
- Recurrent Neural Network (RNN)
  - Processes data in time steps
  - Considers the output of the last step (captures sequential dependencies)